

Alex Liu

haozhouliu17@gmail.com | 352-328-4805 | linkedin.com/in/alex-liu7 | github.com/heimweh17 | U.S. Citizen | **Personal Page:** aliu.me

EDUCATION

University of Florida (UF) - Gainesville, FL

Expected Graduation: May 2028

Bachelor of Science in Computer Science | GIS Certificate

GPA: 3.78/4.0

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Programming Language Concepts, Information & Database Systems

EXPERIENCE

Database Mart | *Operational Engineering Intern - Kansas City, MO*

May 2026 - Jul 2026

- Provisioned **~60-70** customer-configured bare-metal servers over the internship, validating hardware and system stability to deliver production-ready capacity for client workloads.
- Completed **300+** service and maintenance requests on customer infrastructure, restoring or expanding server capacity through hardware upgrades, replacements, and troubleshooting.
- Racked and deployed servers in the data center, configuring host networking and IPMI/BMC management and validating connectivity, hardware health, and system readiness.
- Developed and deployed a FreeDOS-based BIOS update automation system supporting ~30 Supermicro motherboard models, automatically matching firmware and enabling teammates to update outdated BIOS versions safely.

GeoPlan Center, University of Florida | *Research Assistant - Gainesville, FL*

Apr 2026 - Present

- Reviewed and processed Florida traffic crash records in Signal Four Analytics, using police narratives, diagrams, and map context to geocode crashes and validate roadway and location classifications.
- Reduced average crash-record processing time by **~75%** by developing an internal automation-assisted review workflow, increasing throughput while preserving manual validation of final classifications.

PROJECTS

Geography Dashboard / *React, TypeScript, FastAPI, PostgreSQL, AWS, Docker, Leaflet*

Nov 2025

- Built a full-stack geospatial analytics platform using React, TypeScript, FastAPI, PostgreSQL, Leaflet, and AWS to ingest CSV/GeoJSON datasets and generate DBSCAN clusters and density grids for city-scale analysis.
- Designed a decoupled hybrid-cloud architecture separating GIS processing from AI inference; automated Docker/GitHub Actions deployments and implemented cross-service tracing, structured logging, and rate limiting for reliability and cost control.

ScamShield Hub / *React, Node.js, Express, MongoDB, JWT, Resend*

April 2026

- Led development of a full-stack phishing-awareness training platform, building a React SPA and Express/MongoDB REST backend for case-based learning, progress tracking, leaderboards, and social interaction.
- Engineered secure account and administration workflows with JWT authentication, email verification, persistent user and case data, and role-based moderation; deployed the frontend on Vercel & API on Render with MongoDB Atlas.

Ability Bridge / *Python, OpenCV, MediaPipe, PyAutoGUI*

Sep 2025

- Built a webcam-based assistive input system using Python, OpenCV, and MediaPipe FaceMesh for hands-free cursor control and Morse-code typing at ~30 FPS.
- Stabilized real-time head-pose and gesture control using solvePnP, EMA smoothing, hysteresis, and debouncing; implemented eyebrow-click and long-blink commands through PyAutoGUI.

SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, Bash

Software & Web: React, FastAPI, Flask, Node.js, PostgreSQL, REST APIs, SQLAlchemy

Cloud & DevOps: AWS, Docker, Git, GitHub Actions, Linux/Unix

Systems & Infrastructure: Data Center Operations, Server Hardware, IPMI/BMC, BIOS/UEFI, FreeDOS, TCP/IP,

Data, CV & GIS: Pandas, NumPy, OpenCV, MediaPipe, Leaflet, GIS